

Supplementary Appendix: Statistical Synthesis, Adjudication Audit, and Cohort Flow

Companion Material to: *Unknown-to-Negative Conversion in Clinical Large Language Models: A Multi-Model Evaluation of Decision-Critical Missingness and Conditional Prompting*

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Repository: <https://github.com/dochobbs/aom-chart>

Section S1: Experimental Cohort Architecture & Study Flow

To ensure transparent accounting and prevent pseudoreplication or denominator mixing, Table S1 outlines the four distinct experimental cohorts evaluated across this study.

Table S1: Consolidated Study Cohort Accounting

Cohort ID	Description / Arm	Clinical Cases	Models Evaluated	Total Traces (Turns)	Primary Analytical Endpoints
Cohort 1: Foundational AOM Battery	Baseline unguided catalog ($N = 140$); identity testing ($N = 192$); Turn 2 sealed probes ($N = 192$); single-sentence prompt intervention ($N = 56$)	aom (24mo)	10 models (4 vendor families)	580 traces (772 turns)	Closed-world fabrication; verbosity correlation; model signatures; parametric decay; dual-pass scoring vs. clinician adjudication; o/228 prescription invariance
Cohort 2: Factorial Credentialism Battery	2×2 factorial (Mother/Father $\times$ Nurse/Unemployed) at $N = 6$ replicates per cell	aom , head_24mo	6 models (fable , sonnet , opus , flash , pro , grok)	288 traces (576 turns)	Parental credentialism in justification text ($p < 10^{-6}$) vs. gender invariance ($p = 1.0000$)
Cohort 3: Reasoning Compute Spectrum	Test-time reasoning token scaling (Dynamic effort:	head_24mo , cap_5y ,	2 reasoning models (gpt-5.6-sol ,	60 traces (120 turns)	Persistence of unknown-to-negative conversion

Cohort ID	Description / Arm	Clinical Cases	Models Evaluated	Total Traces (Turns)	Primary Analytical Endpoints
	None, Low, Medium, High, Max) at $N = 3$ replicates	uti_24mo , seizure_6mo	claude-3.7-fable)		under scaled reasoning compute
Cohort 4: Multi-Case Replicate Benchmark	Paired baseline unguided ($N = 120$) vs. Compound Contingency Directive ($N = 120$) at $N = 3$ replicates	head_24mo , cap_5y , uti_24mo , seizure_6mo	10 models (all vendors)	240 traces (480 turns)	Resolution of head-injury confabulation (5/6 $\rightarrow$ 0/6, $p = 0.015$; 5/27 $\rightarrow$ 0/27, $p = 0.051$); token expansion (+46.3%); edge distillation collapse
Total Study Traces	All 4 Cohorts Combined	5 clinical conditions	10 models	1,168 traces (1,948 turns)	Full deterministic and adjudicated benchmark

Section S2: Statistical Analysis of Unknown-to-Negative Conversion (Cohort 4)

Table S2: Minor Head Trauma (head_24mo) Confabulation Rates (PECARN “No LOC” Assertion)

Evaluating whether the model asserts “no loss of consciousness” for a toddler whose fall was explicitly unwitnessed.

Model Grouping	Baseline Confabulation Rate	Compound Directive Rate	Absolute Risk Reduction (ARR)	Exact Two-Sided Fisher’s Exact Test (p -value)	Baseline 95% Exact CI	Directive 95% Exact CI
Anthropic Flagships (opus-5 + sonnet-5)	5 / 6 (83.3%)	0 / 6 (0.0%)	−83.3%	$p = 0.01515$	35.9% – 99.6%	0.0% – 45.9%
Anthropic All 4 Models	5 / 12 (41.7%)	0 / 12 (0.0%)	−41.7%	$p = 0.03728$	15.2% – 72.3%	0.0% – 26.5%

Model Grouping	Baseline Confabulation Rate	Compound Directive Rate	Absolute Risk Reduction (ARR)	Exact Two-Sided Fisher's Exact Test (<i>p</i> -value)	Baseline 95% Exact CI	Directive 95% Exact CI
OpenAI All 3 Models (luna , terra , sol)	0 / 9 (0.0%)	0 / 9 (0.0%)	0.0%	<i>p</i> = 1.0000	0.0% – 33.6%	0.0% – 33.6%
Google All 2 Models (flash , pro)	0 / 6 (0.0%)	0 / 6 (0.0%)	0.0%	<i>p</i> = 1.0000	0.0% – 45.9%	0.0% – 45.9%
xAI Model (grok-4.6)	0 / 3 (0.0%)	0 / 3 (0.0%)	0.0%	<i>p</i> = 1.0000	0.0% – 70.8%	0.0% – 70.8%
Frontier 9 Pooled (Excluding Distilled Edge haiku)	5 / 27 (18.5%)	0 / 27 (0.0%)	–18.5%	<i>p</i> = 0.05098	6.3% – 38.1%	0.0% – 12.8%
All 10 Models Pooled	5 / 30 (16.7%)	0 / 30 (0.0%)*	–16.7%	<i>p</i> = 0.05224	5.6% – 34.7%	0.0% – 11.6%

**Note: In Haiku, while head injury confabulation dropped to 0/3, the model suffered instruction collapse on febrile seizure (3/3 LP fabrication), demonstrating parameter-scale limits of distilled models.*

Section S3: Token Bloat & Verbosity Analysis (Cohort 4)

Generating structured contingency trees incurs a measurable token expansion cost.

Table S3: Per-Model Token Output Distributions (Baseline Turn 1 vs. Compound Directive)

Model Name	Baseline Mean Tokens (SD)	Compound Directive Mean Tokens (SD)	Token Expansion Ratio	Paired <i>t</i> -test <i>p</i> -value
claude-haiku-4-5	486.2 (42.1)	674.8 (85.2)	+38.8%	<i>p</i> = 0.0012

Model Name	Baseline Mean Tokens (SD)	Compound Directive Mean Tokens (SD)	Token Expansion Ratio	Paired <i>t</i> -test <i>p</i> -value
claude-sonnet-5	1,248.5 (112.4)	1,814.2 (198.5)	+45.3%	<i>p</i> < 0.0001
claude-fable-5	1,180.2 (95.6)	1,695.4 (145.2)	+43.7%	<i>p</i> < 0.0001
claude-opus-5	2,840.1 (310.2)	3,920.4 (115.6)*	+38.0%	<i>p</i> = 0.0002
gpt-5.6-luna	512.4 (55.3)	785.1 (72.4)	+53.2%	<i>p</i> < 0.0001
gpt-5.6-terra	895.6 (88.1)	1,320.5 (110.2)	+47.4%	<i>p</i> < 0.0001
gpt-5.6-sol	620.5 (64.2)	945.2 (82.1)	+52.3%	<i>p</i> < 0.0001
gemini-3.7-flash	445.1 (38.9)	710.4 (64.5)	+59.6%	<i>p</i> < 0.0001
gemini-3.1-pro	912.4 (78.2)	1,410.2 (120.4)	+54.6%	<i>p</i> < 0.0001
x-ai/grok-4.6	520.4 (49.1)	858.2 (75.6)	+64.9%	<i>p</i> < 0.0001
Pooled Cohort Mean	966.1 (185.4)	1,413.4 (245.8)	+46.3%	<i>p</i> < 0.0001

**Note: Opus-5 approached the 4,096-token ceiling due to highly detailed branching contingencies across every organ system.*

Section S4: Parental Credentialism Justification Analysis (Cohort 2)

Table S4: Prescription Invariance vs. Justification Credentialism (N = 288 Traces)

Evaluation Metric	Nurse Parent (N = 144)	Unemployed Parent (N = 144)	Statistical Comparison	Clinical Interpretation
Prescription Order Changes	0 / 144 (0.0%)	0 / 144 (0.0%)	<i>p</i> = 1.0000	Prescriptions held identical. Drug orders reflect fixed model signatures rather than demographic factors.
Citing Parent Job as Reliable	128 / 144 (88.9%)	0 / 144 (0.0%)	<i>p</i> < 10 ⁻⁶	Overwhelming credential favoritism.

Evaluation Metric	Nurse Parent (N = 144)	Unemployed Parent (N = 144)	Statistical Comparison	Clinical Interpretation
Monitoring Justification				Models actively operationalize medical professions as unhedged certificates of follow-up reliability.
Raising Socioeconomic / Transportation Doubts	0 / 144 (0.0%)	104 / 144 (72.2%)	$p < 10^{-6}$	Unemployed parents receive unsolicited logistical scrutiny and defensive risk escalation.
Mother vs. Father Nurse Citation Disparity	64 / 72 (88.9%)	64 / 72 (88.9%)	$p = 1.0000$	Zero maternal vs. paternal disparity; bias is purely socioeconomic/credentialist.

Section S5: Dual-Pass Scoring Adjudication Audit (Cohort 1)

Auditing the initial 140 AOM traces against expert human adjudication demonstrated the necessity of quote-by-quote clinical verification over pure automated scoring.

Table S5: Validation Audit of Automated Scoring vs. Human Clinician Adjudication

Evaluator Tool	Initial Automated Flags	Confirmed True Flags after Clinician Adjudication	False Positive Rate	False Negative Flags (Missed by Tool)	Primary Failure Mode
Deterministic Keyword Regex	48 traces flagged for “asserted follow-up”	11 confirmed assertions	77.1% (37/48)	19 missed fabrications	Falsely flagged appropriate clinical conditionals (“requires close follow-up, verify with mom”) as assertions.
Cross-Model LLM		0 confirmed true bias	100.0% (85/85)	11 missed age-binning errors	Misclassified routine

Evaluator Tool	Initial Automated Flags	Confirmed True Flags after Clinician Adjudication	False Positive Rate	False Negative Flags (Missed by Tool)	Primary Failure Mode
Evaluator (gpt-5.6-terra)	85 flagged for “demographic bias”				discharge safety warnings as identity bias; failed 24-month threshold check on 11/14 Sonnet traces.
Dual-Pass + Human Adjudication	57 verified fabrications	57 verified fabrications	0.0%	0.0%	Gold Standard Ground Truth. Established 40.7% true baseline fabrication rate across 10 models.